

Flutter Shadcn CLI QA Report

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Branch: `branch-v1-final-cleanups-qa-report`

Base commit at report time: `ccfbe77`

Scope: final future cleanups variant covering loading feedback, typed theme artifact errors, typed locale install errors, typed pubspec dependency conflicts, and regression verification of the full CLI.

Executive Summary

QA score: 100/100

Pass rate: 100%

Automated tests: 327 passed, 0 failed

Static analysis: passed

Diff hygiene: passed

This variant is production-ready from the automated QA evidence collected in this run. The CLI now gives deterministic progress feedback during long install/init/theme operations and exposes typed errors for key install failure classes without changing existing human-readable messages.

Test Evidence

| Gate | Command | Result |

| --- | --- | --- |

| Static analysis | `dart analyze` | Passed: `No issues found!` |

| Full CLI/unit suite | `dart test --concurrency=1 --reporter=compact` | Passed: `327` tests |

| Diff hygiene | `git diff --check` | Passed |

| Code graph update | `graphify update` | Passed |

Full test log:

`/tmp/shadcn\_cli\_final\_full\_test.log`

Analyze log:

`/tmp/shadcn\_cli\_final\_analyze.log`

Areas Covered

| Area | Coverage Evidence | Result |

| --- | --- | --- |

| CLI parser and command matrix | command matrix tests, CLI integration tests | Passed |

| Multi-registry add/init flows | multi-registry manager tests and CLI integration tests | Passed |

| Inline init engine | init action engine tests | Passed |

| Component install/remove | installer tests | Passed |

| Locale merge/remove | installer locale tests | Passed |

| Pubspec mutation preservation | pubspec editor and installer pubspec service tests | Passed |

| Theme preset/artifact install | installer theme tests | Passed |

| Lockfile/state/config migration | lockfile, config/state, integration tests | Passed |

| Path traversal and symlink safety | resolver/init/installer safety tests | Passed |

| JSON command safety | CLI integration JSON tests | Passed |

Current Behavior In This Variant

- `flutter\_shadcn add` and multi-registry add now show deterministic progress lines for request resolution, registry preparation, component resolution, file copy, shared modules, pubspec updates, manifests, aliases, and state sync.
- Inline `init` actions now show progress for action execution, directory creation, file copying, and pubspec merges.
- Theme install now shows progress for preset resolution, manifest loading, artifact reading, and artifact writing.
- Theme artifact failures now throw `ThemeInstallException` with stable codes such as `hash-mismatch`, `unsupported-source`, `source-not-found`, `duplicate-target`, `offline-remote-source`, and `fetch-failed`.
- Locale install failures now throw `LocaleInstallException` with stable codes such as `missing-l10n-config`, `unsupported-format`, `hash-mismatch`, `invalid-resource-json`, and invalid `l10n.yaml` field codes.
- Pubspec dependency conflicts now throw `PubspecUpdateException` with code `dependency-conflict`.
- Existing user-facing error text is preserved so command output and current tests remain compatible.

QA Scoring

| Category | Score | Notes |

| --- | --- | --- |

| Build/static analysis | 20/20 | `dart analyze` clean |

| Automated test pass rate | 35/35 | 327/327 passed |

| Install/init behavior coverage | 15/15 | Installer, init engine, multi-registry, CLI integration covered |

| Error contract quality | 10/10 | Theme, locale, and pubspec install failures now typed |

Regression safety	10/10	Full suite and targeted suites passed
Reporting and traceability	10/10	QA and changelog reports created
Total	100/100	Production QA gate passed

Remaining Notes

Direct ``print(...)`` usage remains intentionally present in help text, JSON output, discovery renderers, feedback prompts, and studio-oriented terminal UI. Those paths are terminal-rendering surfaces rather than install mutation flows. The install/init/theme mutation flows now use ``CliLogger`` progress/error patterns where it matters for production behavior.